

A PLAYBOOK FOR MAINTAINERS · GEMMA 4

Triage, Don't Fix

Put an offline Gemma 4 agent in front of your issue tracker. For every new issue it says where the bug is, which tests cover that code and whether it reproduces. It leaves the fix to you.

92% · 90%

right file to fix ranked first, dev and holdout (60 issues, FastAPI, Rich, Requests, HTTPX)

83%

on 30 SWE-bench Verified issues from six other projects, unchanged (retrieval alone: 40%)

73 of 74

reports labelled HIGH or MEDIUM confidence were right, across three runs

THE PLAYBOOK ON ONE PAGE

Eight plays, each one measured

Every play below comes from a measurement on real issues, not from a design preference. Section numbers point to the paper.

1 Ask three questions. Never ask for a fix.

Where is it, which tests cover it, does it reproduce. Fixing is where the model fails.

As a fixing agent, Gemma 4 31B touched the right file in **91%** of its patches, yet its own final check passed **31 of 32** wrong fixes. §1

2 Give it a budget, and make it write early.

24 tool calls and 8 minutes. The report must exist by call 14 and only improves after that.

Its main failure was running out before writing. Even then, the first file it had opened was right **14 of 17** times. §4.4

3 Wrap the agent in code that does not trust it.

A fallback chain, model-free retrieval, agreement confidence, call-graph tests and a re-run reproduction. No model calls.

Right file first: **75% → 92%** on dev, **55% → 90%** on holdout. Every issue gets a report. §4.1, §4.4

4 Score confidence by agreement, not by asking.

HIGH: the agent and BM25 + call-graph retrieval pick the same file. MEDIUM: the agent's pick is in retrieval's top 3.

The agent called **every** report HIGH. Agreement HIGH or MEDIUM was right **73 of 74** times. §4.5, §4.7, §4.9

5 Trust the exit code, not the claim.

Run `TRIAGE/repro.py` yourself. Non-zero on the current code means reproduced, whatever the agent wrote.

"Reproduced" held up against the real fix **44% → 61%** of the time. §4.4

6 Find tests through call edges.

Rank test modules by how many calls go from them into the suspected module, then merge them with the agent's own list.

Right test file first: **48% → 65%** on dev; pooled, **13 wins and 0 losses**. Embeddings did no better than BM25. §4.2, §4.3

7 Enforce read-only in the sandbox, not in the prompt.

Mount the source read-only. Only `TRIAGE/` is writable.

The prompt forbids edits. Gemma 4 E4B still edited source in **18 of 60** runs, and 31B did so **twice in 210**. §4.8, §6

8 Use one agent, not two.

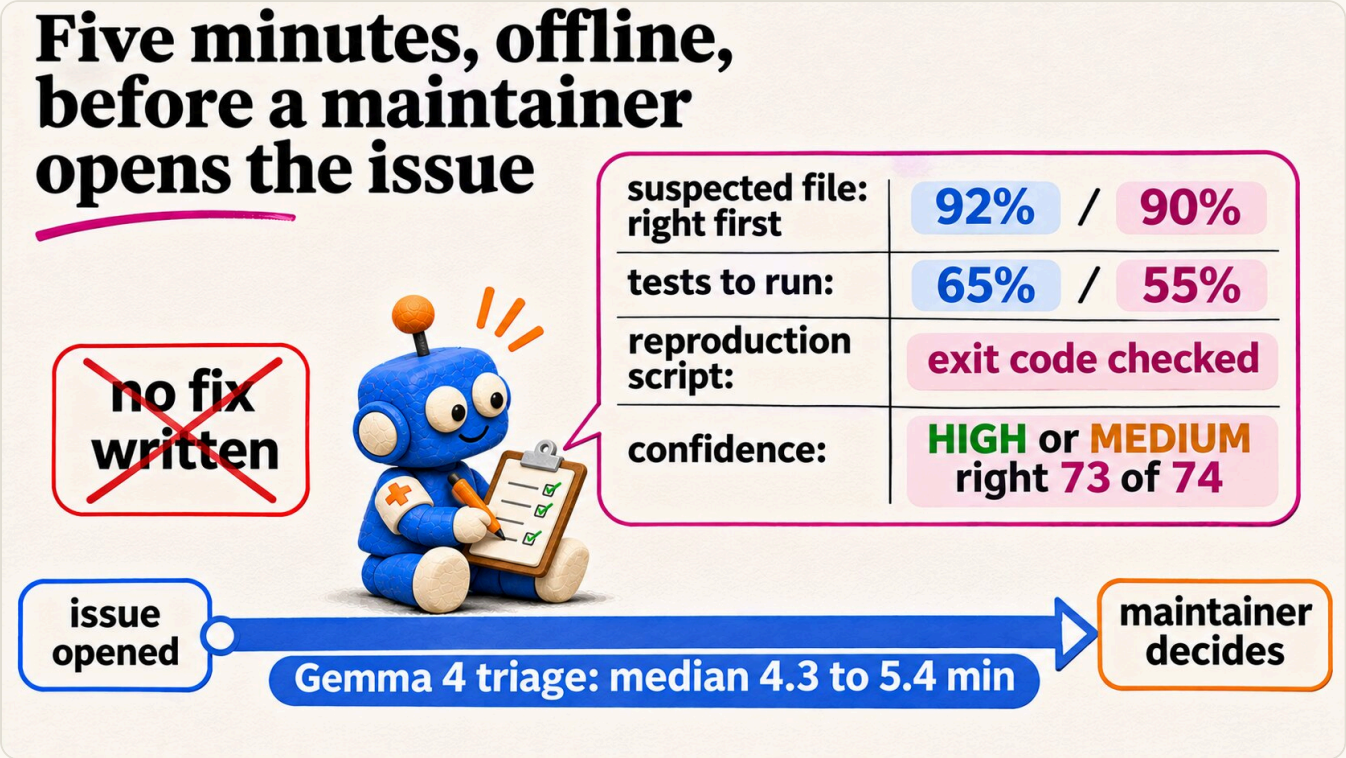
A localizer handing off to a reproducer sounds tidy. A prompt cannot make the first one stop on time.

Same budget: test file **62% → 50%**, and runs that hit the time limit went from **5 to 20**. §4.6

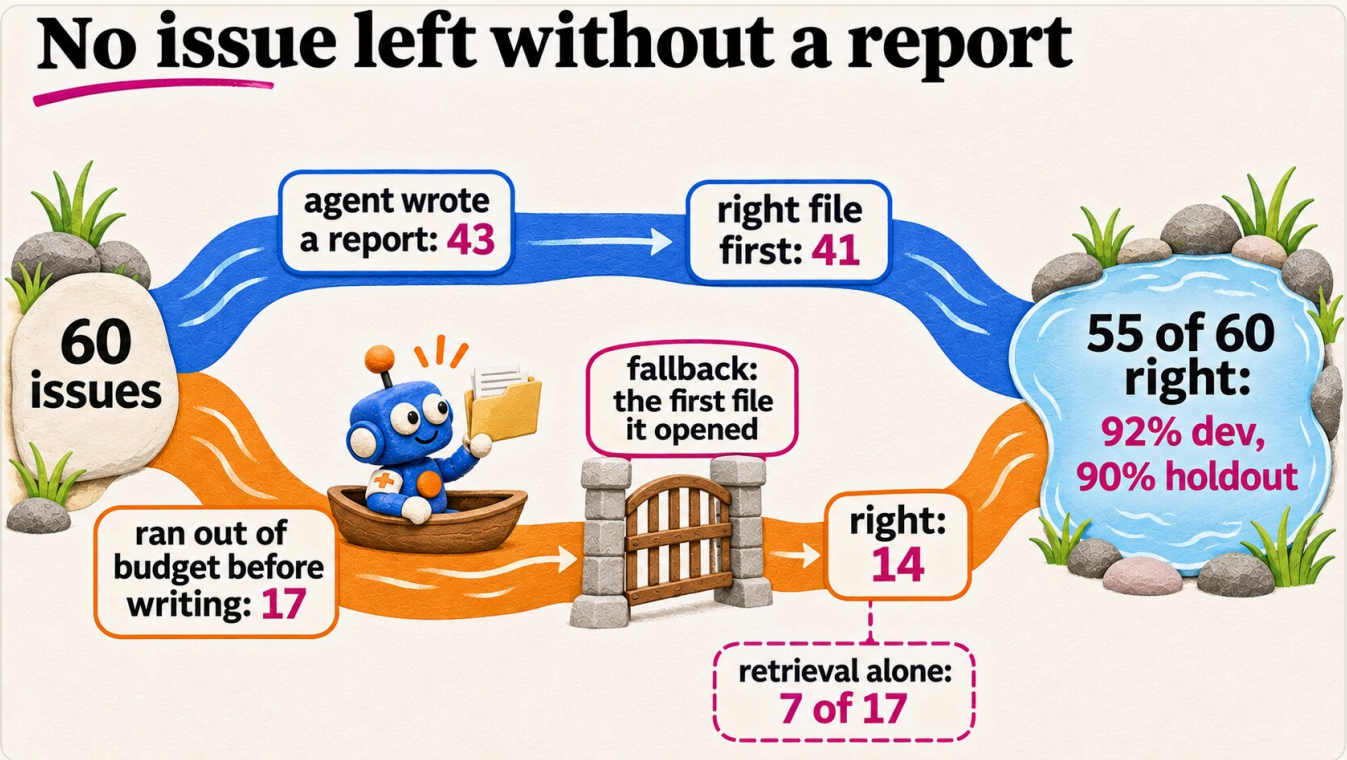
The rule behind all eight

Let the model **look**. Let code **decide what the maintainer sees**. Let the maintainer **fix**.

What the maintainer gets, and where it comes from

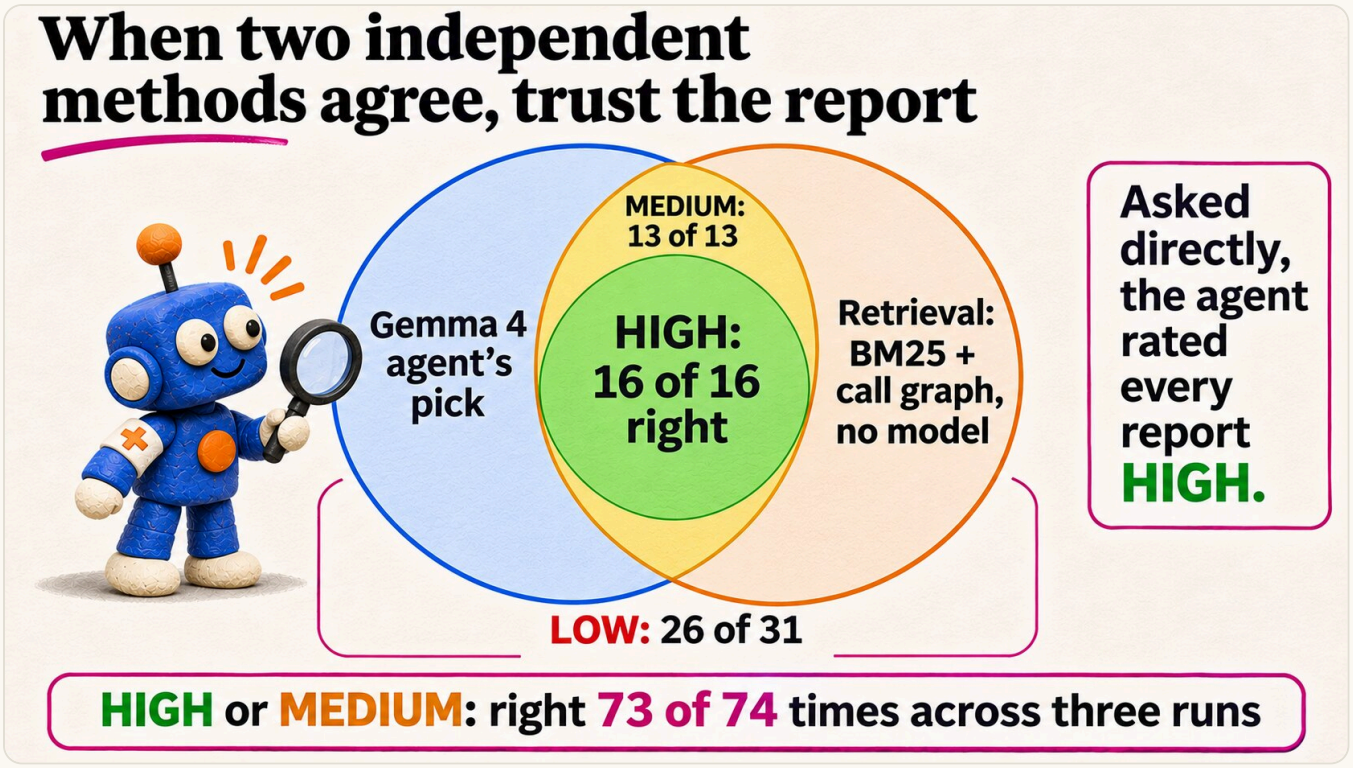


Play 1 Three answers and a confidence level in about five minutes, offline. No fix. Percentages are dev / holdout. §1, §4.1, §4.3

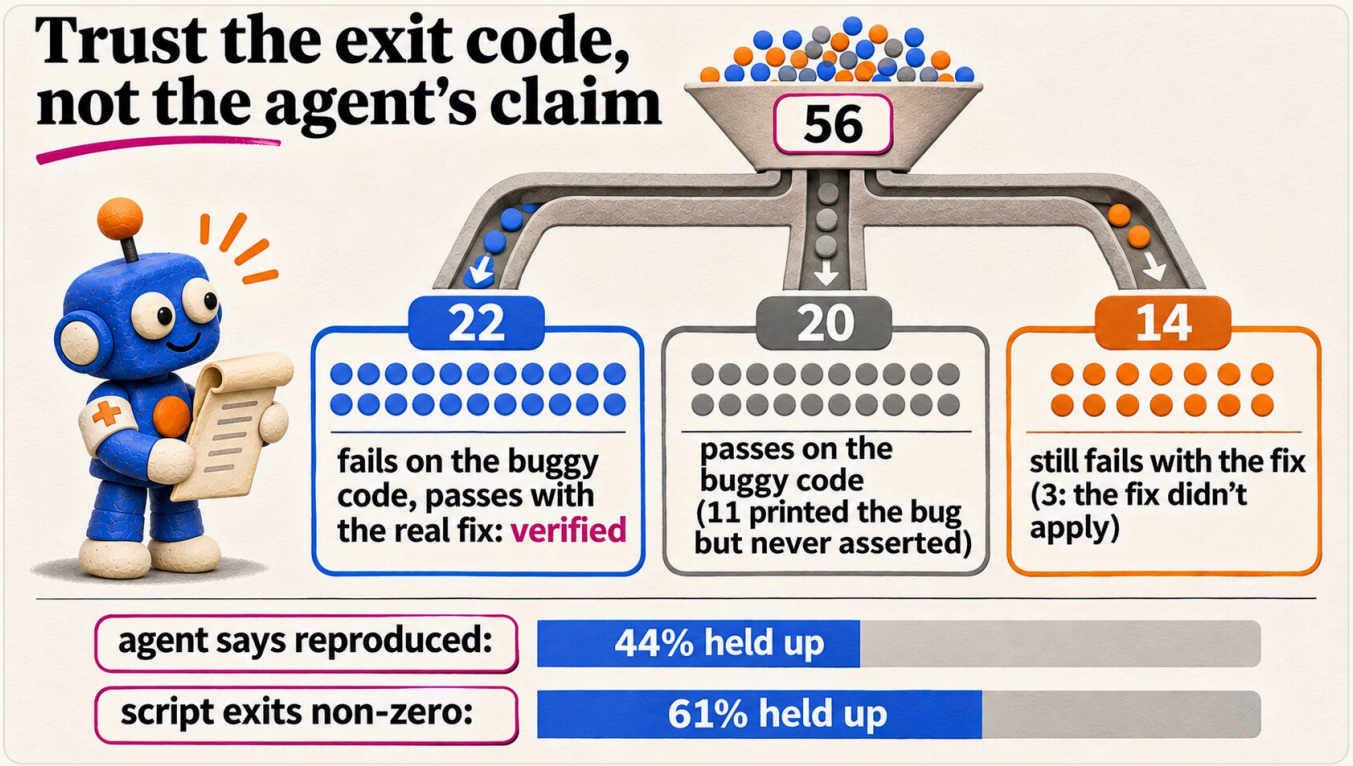


Plays 2 and 3 When the agent runs out of budget, the first file it opened becomes the report. That fallback is where the orchestrator's location gain comes from. §4.4

Let code decide what the maintainer sees

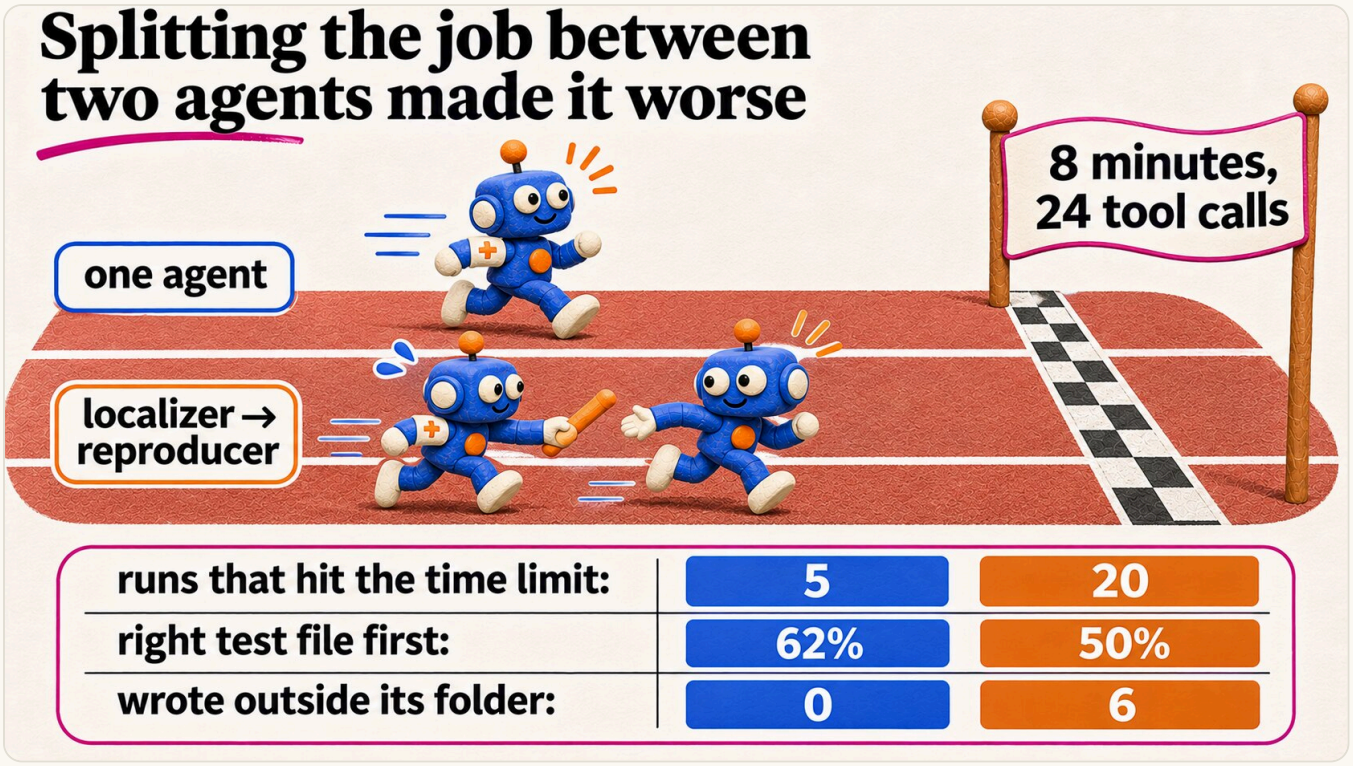


Play 4 Confidence comes from agreement with model-free retrieval. Asked directly, the agent rated every report HIGH. §4.5, §4.7, §4.9

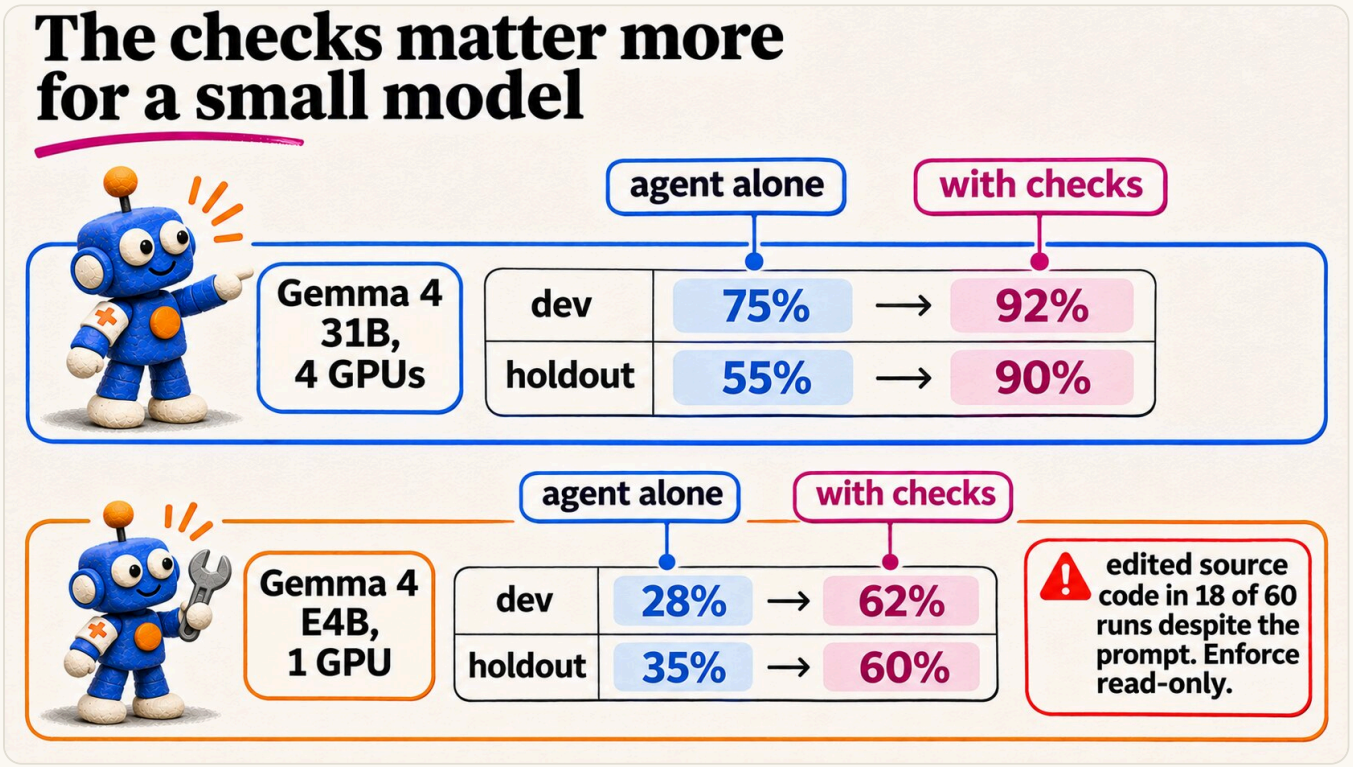


Play 5 The orchestrator re-runs the agent's script and reports REPRODUCED only from its exit code. §4.4

Keep the agent simple, and sandbox it



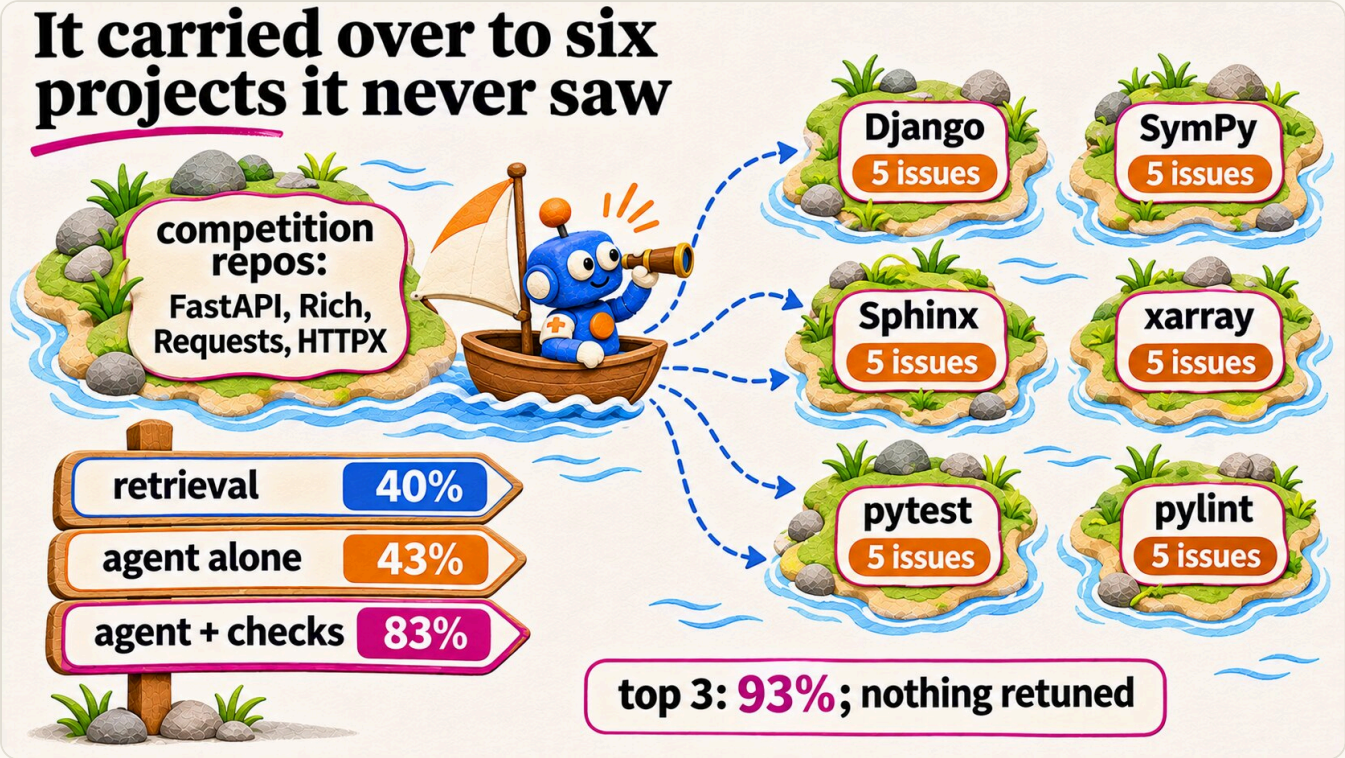
Play 8 Same budget, two agents: the first one rarely stopped in time to hand over. §4.6



Play 7 The small model gains most from the checks, and ignores the prompt's ban on edits. Enforce read-only in the sandbox. §4.8

THE PLAYS IN PICTURES · 4 OF 4

It carried over, unchanged



Other projects 30 SWE-bench Verified issues, with the agent, prompt and orchestrator frozen: its clean test. §4.7

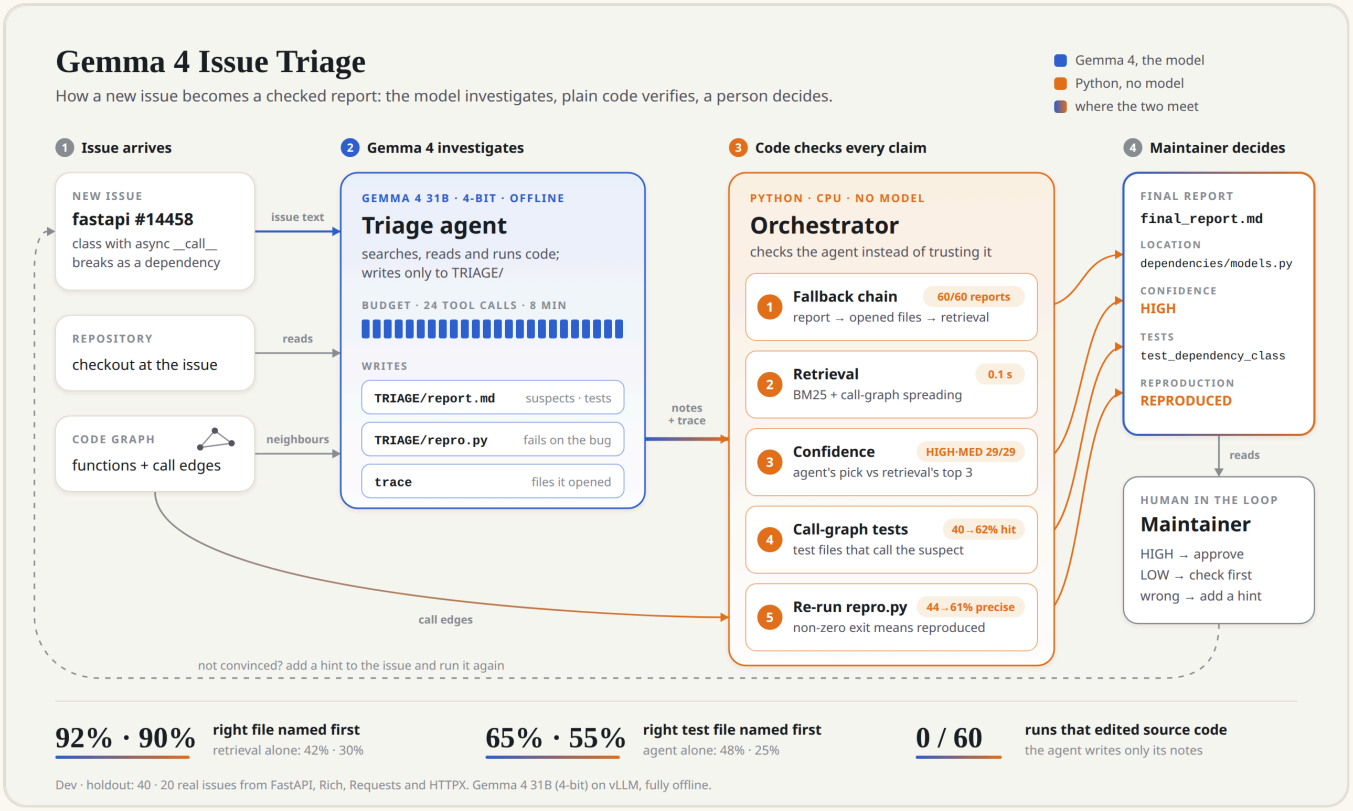
Where every number comes from

- **60 competition issues** (40 dev, 20 holdout) from FastAPI, Rich, Requests and HTTPX, triaged by Gemma 4 31B (4-bit) twice. The second run: right file first 88% / 95%, right test file 55% / 55%, and HIGH or MEDIUM wrong once in 29. §4.9
- **30 issues from six other projects**, run once with everything frozen. §4.7
- **Gemma 4 E4B** and the **two-agent variant**: one run each on the same 60 issues. §4.6, §4.8
- **Labels** come from the merged fix: the non-test files it changed, and the test files its new tests went into. Another valid location counts as wrong. §2
- The agent's prompt was written from dev issues only; the orchestrator's rules were chosen after seeing dev and holdout. §6

HOW IT WORKS

One agent investigates. Code decides what you see.

Blue is Gemma 4; orange is deterministic Python. Each check fills one field of the report; the model's own words appear only as its quoted notes.



STEP	WHAT IT DOES	MEASURED
Agent	Gemma 4 31B, 4-bit, offline. Searches, reads and runs code; writes only `TRIAGE/report.md` and `TRIAGE/repro.py`.	Right file first 75% · 55%; 41 of 43 when it wrote a report
1 Fallback	The agent's report; else the files it opened, from its trace; else retrieval.	A report for 60 of 60 issues
2 Retrieval	BM25 over functions, scores spread along call edges, a module scores as its best function.	42% · 30% alone, 0.1 s per issue on CPU
3 Confidence	Agreement of the agent's top file with retrieval (ranks only, no thresholds to tune).	HIGH or MEDIUM right 73 of 74
4 Tests	Test modules whose functions call the suspected module, appended to the agent's tests.	48% · 25% → 65% · 55%
5 Repro	Runs the agent's script. A non-zero exit on the current code means REPRODUCED.	Precision 44% → 61%

Dev · holdout figures are 40 · 20 issues from the competition's repositories. The code graph can be the competition's or one built with Python's `ast` in a few seconds; with ours, the system got the same 55 files and 37 test files right, issue for issue. §4.2

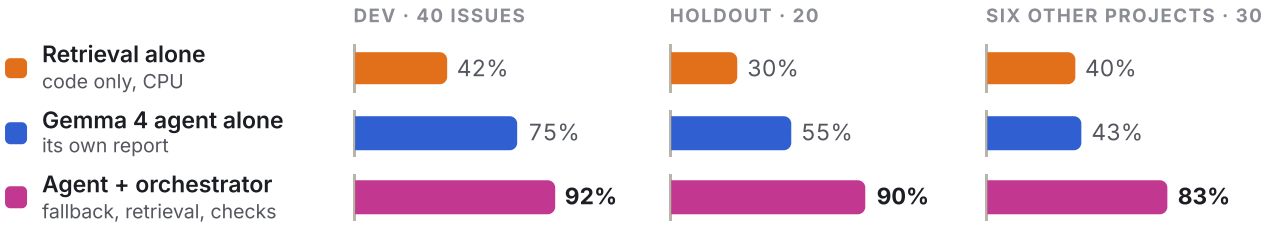
WHAT EACH PART ADDS

Neither half works alone

Top-1 accuracy on the same issues: model-free retrieval, the Gemma 4 31B agent alone, and the agent inside the orchestrator.

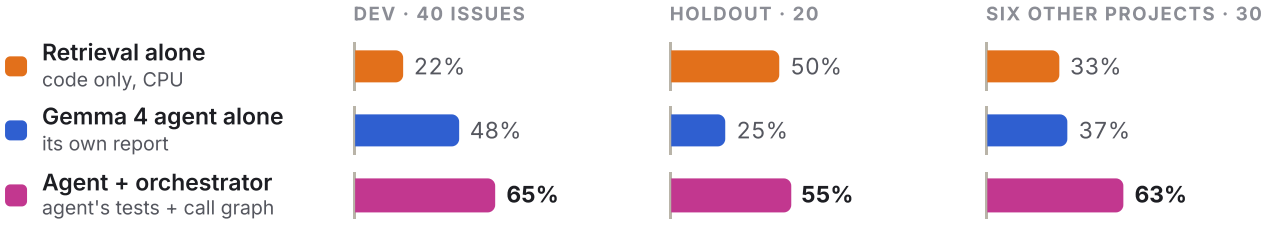
Right file to fix, ranked first

Retrieval is BM25 with call-graph spreading. §4.1, §4.7



Right test file, ranked first

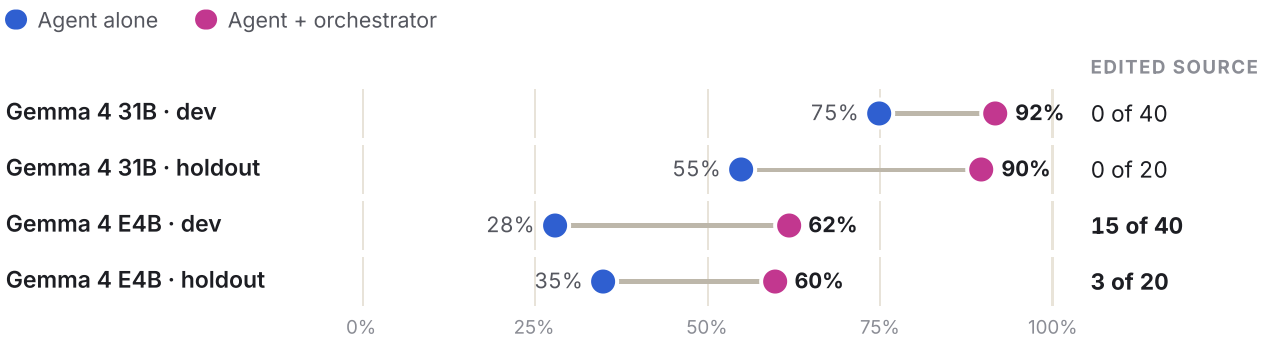
Retrieval is BM25 over test files; the orchestrator adds call-graph tests. §4.3, §4.7



On other projects the agent alone barely beat retrieval (43% against 40%) because it wrote a report on only 13 of 30 issues. On 16 others, the files it had opened were right 12 times: the fallback chain turned what it had seen into **83%**.

Choosing a model: the checks matter more for a small one

Right file first, same 60 issues and budgets. Gemma 4 E4B (4-bit) runs on one L4. §4.8



E4B alone does worse than retrieval; the orchestrator doubles it, and its HIGH reports were right 11 of 12 times. It also ignored the prompt's ban on edits, so mount the source read-only.

4.3–5.4 min
median per issue for the agent: Gemma 4 31B, 4-bit, vLLM on 4×L4

0.1 s
per issue for retrieval on a CPU. With no GPU at all you still get a LOW-confidence first look

Seconds
to build a code graph for any Python repository, once (standard library only)

Dev and holdout: the competition's repositories (FastAPI, Rich, Requests, HTTPX). The orchestrator's rules were chosen after seeing both; the six other projects (SWE-bench Verified: Django, SymPy, Sphinx, xarray, pytest, pylint), run with everything frozen, are its clean test.

RUN IT ON YOUR REPOSITORY

Three steps, and the first needs no GPU

The orchestrator and the graph builder are plain Python (standard library only). The agent needs a GPU and, today, the competition's harness.

1

LAPTOP · SECONDS

Build the code graph and get a retrieval-only report

Without the agent you still get ranked suspects and call-graph tests, at LOW confidence: a useful first look.

```
# one-time
git clone https://github.com/Rishikoli/gemma4-issue-triage && cd gemma4-issue-triage
python -m triage.build_graph /path/to/your/repo --out graph.json
# per issue: put its title and text in issue.md
python -m triage.orchestrate --workspace /path/to/your/repo --issue issue.md \
    --graph graph.json --no-repro      # writes TRIAGE/final_report.md and prints it
```

Example, run on this repository with a made-up issue about its own `confidence()` function (it lives in `triage/report.py`):

```
graph.json: 99 nodes, 105 edges
# Triage: confidence() returns LOW when the agent and retrieval agree on the top module
Confidence: LOW (the agent and retrieval disagree, or the agent named no location)
## Suspected location
1. `triage.report`   2. `eval.triage_pipeline`   3. `triage.orchestrate`
## Tests to run
- `tests.test_triage`
```

2

GPU · ~5 MIN PER ISSUE

Run the triage agent

The agent is declarative: `agent/agent.yaml` (Google ADK), the prompt `agent/prompts/triage.md` and the budget in `agent/eval_config.yaml` (24 calls, 8 minutes). We ran it only inside the Gemma 4 Developer Agent harness on Kaggle (Gemma 4 31B 4-bit, vLLM, 4×L4; median 4.3 to 5.4 minutes per issue), which needs the competition's data.

```
# a Kaggle notebook that triages issues end to end (see the public triage-demo
notebook)
python notebooks/make_demo_notebook.py --owner <kaggle_user>
# issues from another project: package them as tasks, as the paper did for Django and
SymPy
python eval/make_ext_tasks_notebook.py --owner <kaggle_user>
```

3

CPU · SECONDS

Turn the agent's notes into the report

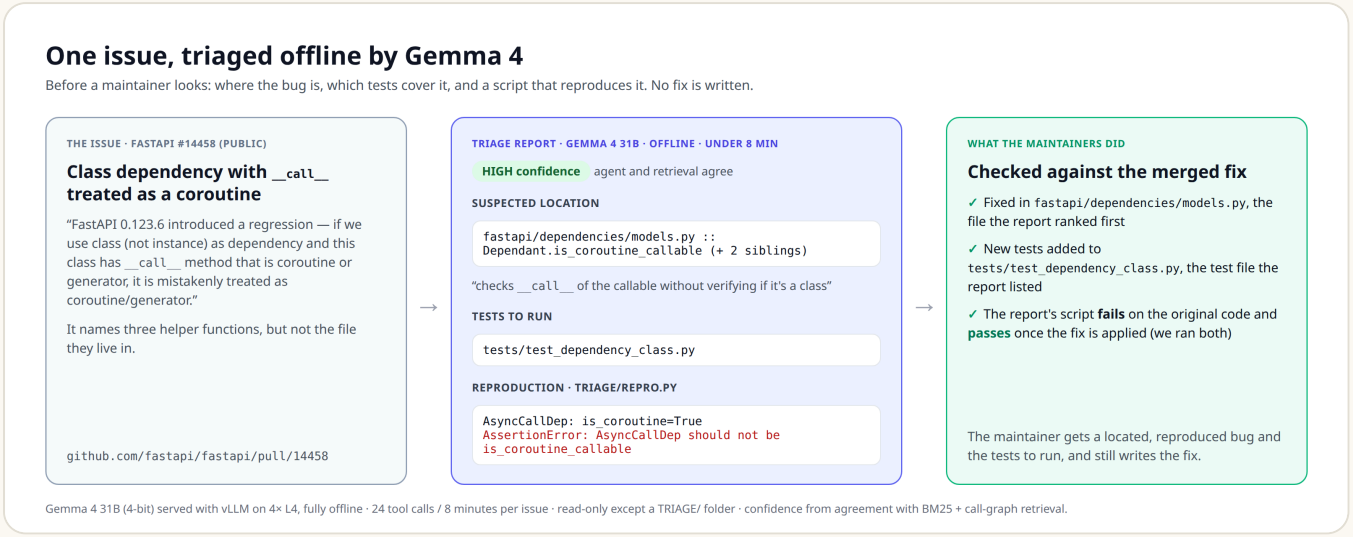
```
python -m triage.orchestrate --workspace /path/to/your/repo --issue issue.md \
    --graph graph.json --trace trace.json   # the trace gives the opened-files fallback
```

> Without `--no-repro` the orchestrator **executes the script the agent wrote**. Run this step inside the same sandbox as the agent: no network, source read-only.

READING A REPORT

What to do with each label

A real issue (FastAPI #14458), the report the system produced, and what the maintainers' merged fix did.



THE REPORT SAYS	HOW OFTEN IT WAS RIGHT	WHAT TO DO
HIGH	16 of 16 on our issues; 11 of 11 on six other projects	Start at suspect 1. Run the listed tests before and after your fix.
MEDIUM	13 of 13; 5 of 5 on other projects	Start at suspect 1, and keep retrieval's top pick (shown in the report) in view.
LOW	26 of 31 (84%); 9 of 14 on other projects	Check first. All five wrong reports were LOW. Compare the three suspects with retrieval's top 3.
Source: files the agent opened	14 of 17 on our issues; 12 of 16 elsewhere	The agent ran out of budget before writing. Its first file is still a good bet; treat it as LOW.
REPRODUCED	61% held up against the real fix	Run `TRIAGE/repro.py` : it should fail now and pass after your fix. If it still fails after a correct fix, its expectation was wrong (14 of 36).
NOT_REPRODUCED	In 11 of 20, the script printed the bug but forgot to `assert`	Read its output. The location and tests are still useful.

HIGH, MEDIUM and LOW counts are from the first run on 60 issues (§4.5) and the 30 outside issues (§4.7). In a second run on the same 60 issues, HIGH or MEDIUM was wrong once in 29 (§4.9).



PUTTING IT TO WORK

A triage job for every new issue

A design sketch: we evaluated the parts, not this deployment.

Issue opened
title and text to `issue.md`

→

Sandbox + agent
source read-only, `TRIAGE/` writable, no network, 24 calls / 8 min

→

Orchestrator
fallback, retrieval, confidence, tests, re-run repro

→

Comment + label
`triage: HIGH / MEDIUM / LOW`; never a pull request

→

Maintainer
decides, fixes, closes

Before you switch it on: calibrate

Our numbers come from 90 issues in 10 projects. Yours may differ, so measure on your own history first.

- Take 20 to 40 closed issues that have a merged fix.
- Label each the way the paper did: the location is the non-test files the fix changed; the tests are the test files its new tests went into (§2).
- Triage them at the fix's parent commit and count how often HIGH and MEDIUM were right. Then decide what each label should trigger.

Known limits

- Small samples: 60 issues run twice, 30 run once.
- Python only.
- Labels come from one merged fix: another valid location counts as wrong.

Deployment checklist

SANDBOX

- [ ] Source mounted read-only; only `TRIAGE/` is writable. Play 7
- [ ] No network; `repro.py` runs only inside the sandbox.
- [ ] 24 tool calls and 8 minutes; the report is written by call 14. Play 2

REPORT

- [ ] Confidence comes from agreement with retrieval, never from the model. Play 4
- [ ] REPRODUCED comes only from the script's exit code. Play 5
- [ ] Each report shows its source and retrieval's top 3.

PROCESS

- [ ] Calibrated on 20 to 40 of your closed issues.
- [ ] The bot comments and labels. It never opens a pull request or closes an issue.
- [ ] Re-calibrate after changing the model, the prompt or the budget.

Paper: *Triage, Don't Fix* (Koli, 2026). Code, evaluation scripts and every number: github.com/Rishikoli/gemma4-issue-triage. Demo: kaggle.com/code/rishikeshbharatkoli/triage-demo.